

Week 1 demo — fallback captures

Library kiosk: one architect-critic cycle on a class diagram

Captured with Claude Code, Sonnet 5 (low effort), September
2026

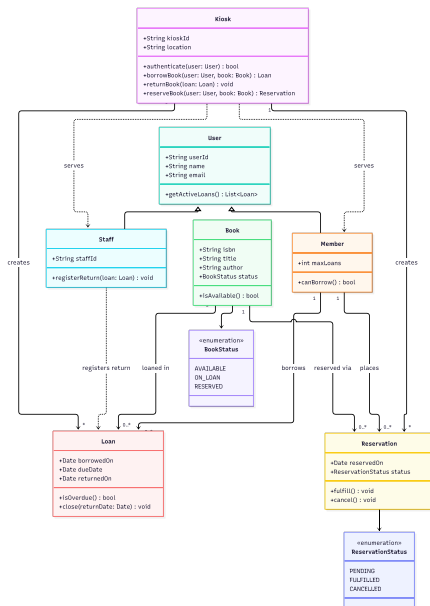
When to use this deck

The live AI is not answering, so here is what the same prompts produced in a real run. We read and critique these diagrams exactly as we would have done live.

Prompt #1

“Generate a UML class diagram in Mermaid for a small library kiosk. Users can borrow and return books. Staff can register returns. A book can be reserved while it is out on loan.”

AI's answer — the diagram



AI's answer — what it said

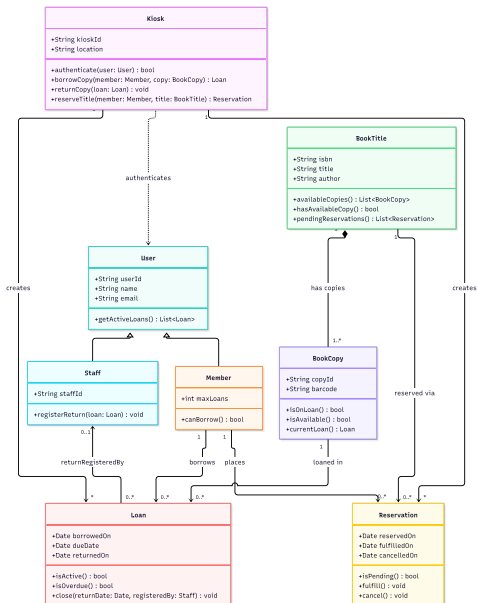
Design notes

- ▶ Member and Staff specialize User. Members borrow and reserve. Staff register returns through `registerReturn`.
- ▶ Loan links a Member to a Book. It stays open until `returnedOn` is set.
- ▶ A Reservation is only valid while its Book is `ON_LOAN`. The kiosk should enforce this in `reserveBook`. When the book is returned, the oldest `PENDING` reservation is fulfilled and the book becomes `RESERVED`.
- ▶ The returns operation is on both `Kiosk.returnBook` (self-service) and `Staff.registerReturn` (staff-registered). If only staff should handle returns, remove `returnBook` from `Kiosk`.

Prompt #2

“Update the diagram: separate a book title from its physical copies — loans are of copies, reservations are of titles; remove the BookStatus enum and derive availability from loans and reservations; make Staff a real participant — link it to the loans whose return it registers.”

AI's revised answer — the diagram



Al's revised answer — what it said

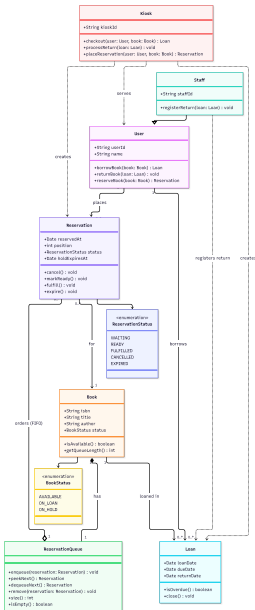
Design notes [...]

- ▶ **No status enum:** state is derived.
 - ▶ A copy is on loan if it has a `Loan` with `returnedOn == null`, and available otherwise.
 - ▶ A copy is held for a reservation if the title has a pending `Reservation` (no `fulfilledOn` or `cancelledOn`) and the copy is not on loan.
 - ▶ `ReservationStatus` is removed for the same reason. `isPending()` checks the dates.
- ▶ **Staff link:** the `Staff` to `Loan` association is `returnRegisteredBy`, optional (0..1) because a self-service return at the kiosk has no staff member. `Loan.close` records who registered the return, and `Staff.registerReturn` calls it.
- ▶ **Open question:** `Kiosk.returnCopy` still allows self-service returns. If only staff should handle returns, remove it and make `registeredBy` mandatory (1).

Reserve prompt — if the first diagram is too clean

“Now redo this for a system where books can be reserved by multiple users in a queue.”

Reserve answer — the diagram



Reserve answer — what it said

What changed

- ▶ `ReservationQueue` is new. Each `Book` owns exactly one, and it holds many `Reservations` in first-in-first-out order.
- ▶ `Reservation` gains `position`, plus a `READY` state and `holdExpiresAt`. [...]
- ▶ `BookStatus` now has `ON_HOLD`, replacing `RESERVED`. A book can have a queue while `ON_LOAN`, so the queue length carries the reservation information and the status doesn't have to.

Business rules the diagram doesn't enforce

- ▶ A user may have only one active reservation per book.
- ▶ Only the user at the head of the queue may borrow an `ON_HOLD` book.
- ▶ `position` is derived from queue order and must be updated when a reservation is cancelled or expires.